MFE FORMWORK MR11 SYSTEM

Data Models

Version 1.0

Prepared for

MFE Formwork

Prepared by

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1. Introduction

This document describes how the MR11 System structures its data. Data passes through six layers on its way from a department's Excel file to what the user's browser remembers, and each layer has its own shape. The sections below take them in that order and explain every field.

The full column-by-column data dictionary for the database, with data types, is in Appendix A of the System Architecture Document. This document focuses on how the pieces relate to one another and what the fields mean.

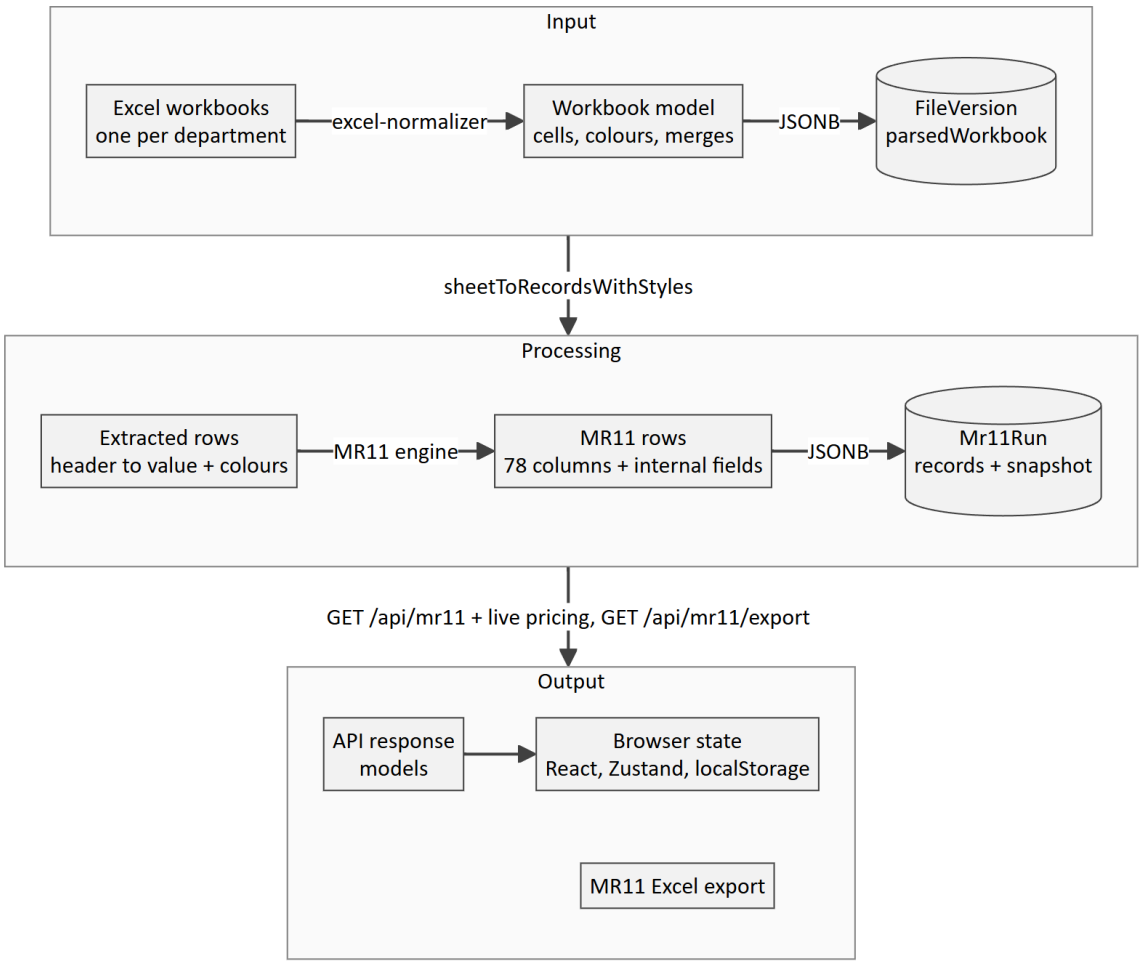


Figure 1: How data moves through the layers of the system

Layer	What it is	Where it lives	Section
Workbook model	Each uploaded sheet as cells, with their values, colours and merges	FileVersion.parsedWorkbook (JSONB)	3
Database tables	Users, roles, departments, workbook versions and MR11 runs	Supabase PostgreSQL	4
Extracted rows	A sheet's rows as header and value pairs, with each row's font and fill colour	Server memory, while an MR11 is being built	5

Layer	What it is	Where it lives	Section
MR11 run and rows	The finished schedule and the history it carries forward	Mr11Run (JSONB)	6
API models	The JSON that the API sends and receives	HTTP requests and responses	7
Browser state	Who is signed in, page settings and filters	React state, Zustand and localStorage	8

2. Conventions used in this document

- Field names are written exactly as they appear in the code (camelCase), and MR11 column names as they appear on screen.
- Dates inside JSON are text in the form YYYY-MM-DD. Database timestamps are TIMESTAMP(3), which is accurate to the millisecond.
- Colours are hex text such as #FF9900, and an empty fill means the cell has no fill.
- Where we say "map", we mean a JSON object used as a lookup, such as one that gives the m² for each month.

3. The workbook model

When a workbook is uploaded, `server/utils/excel-normalizer.ts` turns each of its sheets into the structure below, and the department page draws the sheet back from it. Dispatch keeps its Local and Overseas files together in a single object.

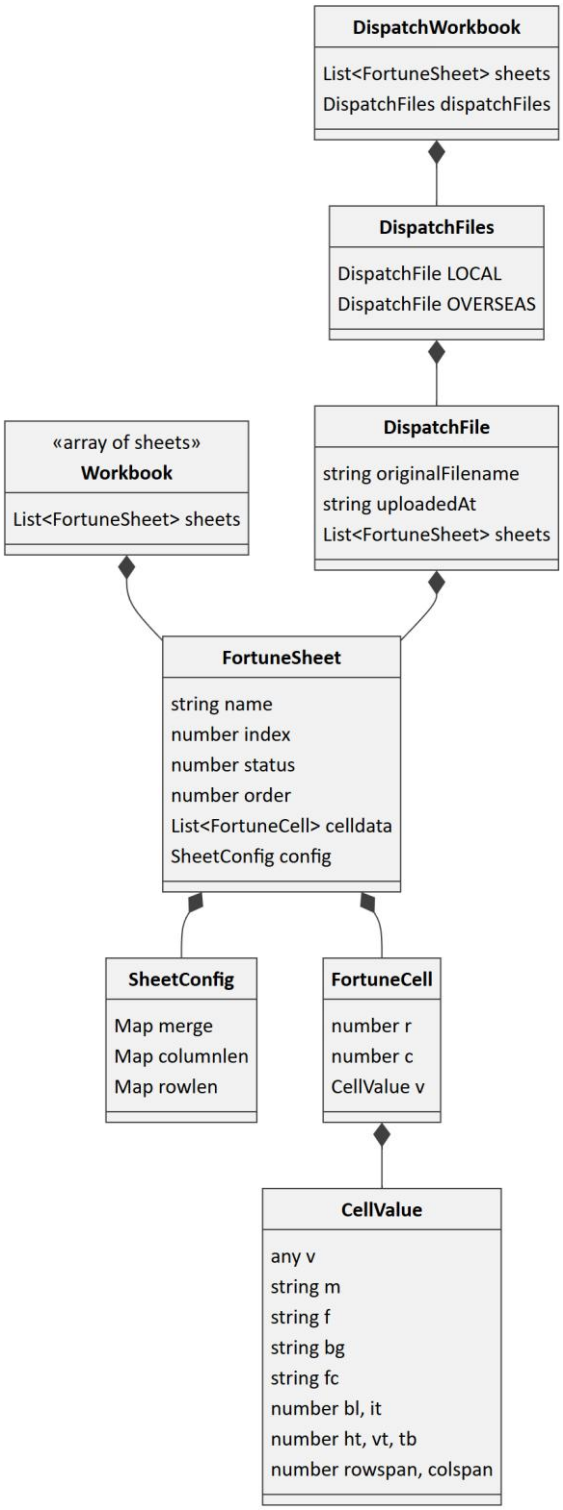


Figure 2: The workbook model stored in FileVersion.parsedWorkbook

3.1 FortuneSheet, one worksheet

Field	Type	What it means
name	Text	The sheet's name. For Dispatch, "Local - " or "Overseas - " is added in front
index and order	Number	The position of the sheet, counting from 0
status	Number	1 for the first sheet and 0 for the others
celldata	A list of FortuneCell	Every cell that is not empty
config.merge	A map from "row_col" to { r, c, rs, cs }	Merged ranges, given by the top-left cell, the number of rows and the number of columns
config.columnlen	A map from column to number	Column widths in pixels (Excel's width multiplied by 8)
config.rowlen	A map from row to number	Row heights in pixels (Excel's height multiplied by 1.33)

3.2 FortuneCell and CellValue, one cell

Field	Type	What it means
r and c	Number	The row and column, counting from 0
v.v	Text, number, true or false, or a date	The value itself. For a formula, this is its calculated result. Every cell inside a merge repeats the top-left value
v.m	Text	The value as display text
v.f	Text	The formula, if there is one, such as =SUM(B2:B9)
v.bg	Text	The fill colour in hex, with theme colours worked out including their tint
v.fc	Text	The font colour in hex
v.bl and v.it	0 or 1	Bold and italic
v.ht	0, 1 or 2	Horizontal alignment: centre, left or right
v.vt	0, 1 or 2	Vertical alignment: middle, top or bottom
v.tb	0 or 2	2 when the text wraps
v.rowspan and v.colspan	Number	The size of the merge, recorded only on its top-left cell

3.3 The Dispatch workbook

Field	Type	What it means
sheets	A list of FortuneSheet	The sheets of both files together, for anything that just needs a simple list
dispatchFiles.LOCAL and dispatchFiles.OVERSEAS	{ originalFilename, uploadedAt, sheets }	Each file on its own, so that an upload replaces only its own part

4. Database tables

Supabase holds eleven tables. Version 1.0 uses six of them. The other five are defined but not written to, because the data they were meant for is kept inside each Mr11Run instead. In the diagrams, "DOUBLE" stands for DOUBLE PRECISION and "TEXT_ARRAY" for TEXT[].

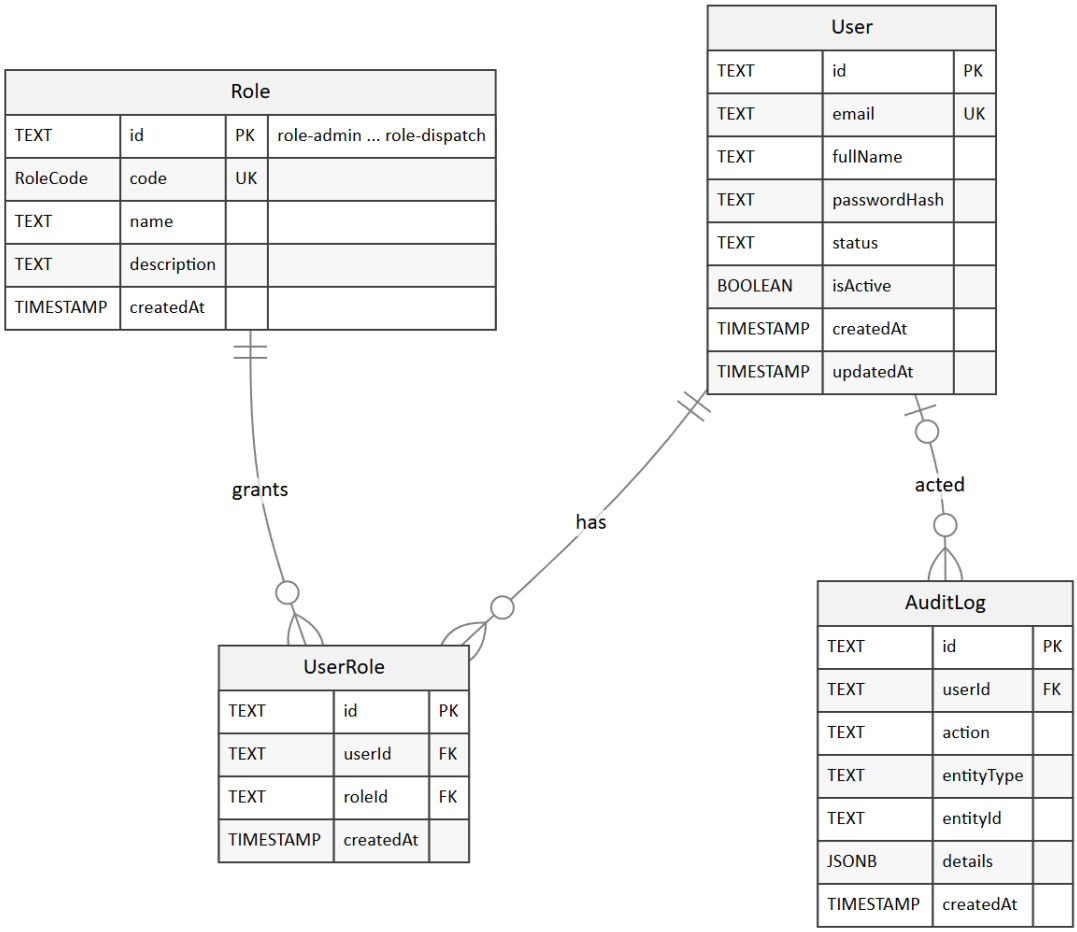


Figure 3: Accounts, made up of users, roles and the audit log

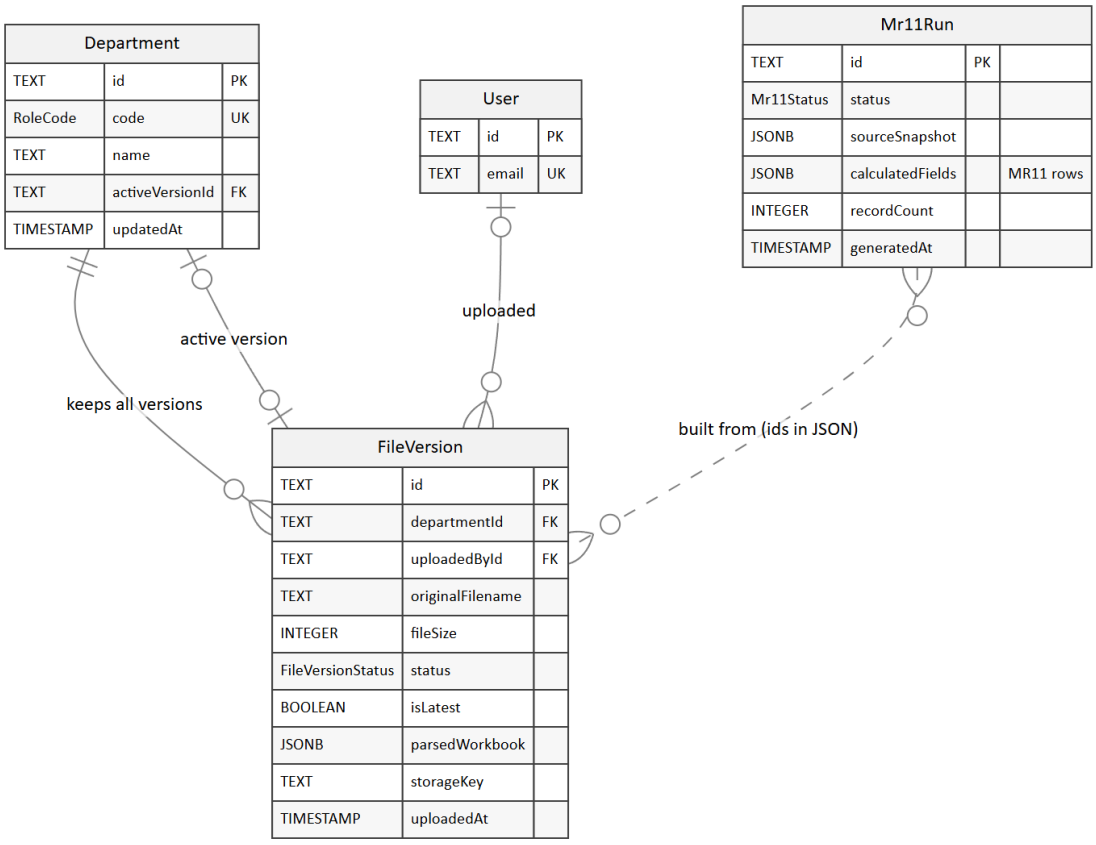


Figure 4: Workbooks and MR11 runs

PlanningSeriesHistory			
TEXT	id	PK	
TEXT	projectNo		unique with stream, fontColor, seriesNumber
TEXT	projectShortname		
TEXT	stream		
TEXT	fontColor		
INTEGER	seriesNumber		
DOUBLE	totalProcessed		

PlanningProjectQuantityTracker			
TEXT	id	PK	
TEXT	projectNo		unique with stream, fontColor
TEXT	stream		
TEXT	fontColor		
DOUBLE	totalTargetQuantity		
DOUBLE	totalProcessed		
DOUBLE	balanceQuantity		

Figure 5: History tables that are defined but not written to in version 1.0 (part 1)

ProductionSeriesHistory			
TEXT	id	PK	
TEXT	projectShortname		unique with stream, fontColor, seriesNumber
TEXT	stream		
TEXT	fontColor		
INTEGER	seriesNumber		
DOUBLE	totalFabricated		

Mr11Config			
TEXT	id	PK	singleton
TEXT_ARRAY	visibleColumns		
TIMESTAMP	updatedAt		

Figure 6: History tables that are defined but not written to in version 1.0 (part 2)

Table	What it holds	It grows with	Used in version 1.0
User	The people who can sign in	New accounts	Yes
Role	The nine roles, with fixed ids	Never	Yes
UserRole	Which user has which role	Role changes	Yes
Department	The seven departments and their active workbook	Never	Yes
FileVersion	Every workbook ever uploaded	Every upload	Yes
Mr11Run	Every build of the MR11	Every upload, sync, row delete or restore	Yes
AuditLog	Records of uploads	Nothing	No; the log is kept in memory
Mr11Config	The visible-column setting	Nothing	No; it is only seeded
PlanningSeriesHistory, PlanningProjectQuantityTracker and ProductionSeriesHistory	Series history	Nothing	No; this is kept in Mr11Run.sourceSnapshot

The enumerated types are RoleCode (ADMIN, CEO, BD, FINANCE, SHELLPLAN, DESIGN, PLANNING, PRODUCTION and DISPATCH), Mr11Status (EMPTY, PARTIAL, READY and FAILED) and FileVersionStatus (UPLOADING, PROCESSING, READY and FAILED).

5. Extracted rows, used while the MR11 is built

The engine's `sheetToRecordsWithStyles()` function reads the first sheet of a workbook into rows. It takes the headers from the top rows, brings up sub-headers that sit under a repeated group header such as "Payment terms", and renames duplicate headers to "Type", "Type 2", "Type 3" and so on.

Field	Type	What it means
data	A map from header to value	The row's values, looked up by header name

Field	Type	What it means
fontColor	Text	The row's font colour, used to match blocks across departments
fillColor	Text	The row's fill colour
rawCells	A map from column to value	The values by column position, used to drop BD rows that are empty
mergedCopyHeaders	A list of text	The headers whose value was copied from a merged cell, so that totals count it only once
dispatchPart	LOCAL or OVERSEAS	For Dispatch only, which file the row came from

6. The MR11 run and its rows

Each build of the MR11 is saved as one Mr11Run. Its rows are the schedule that users see, and its snapshot carries forward everything the next build needs to know.

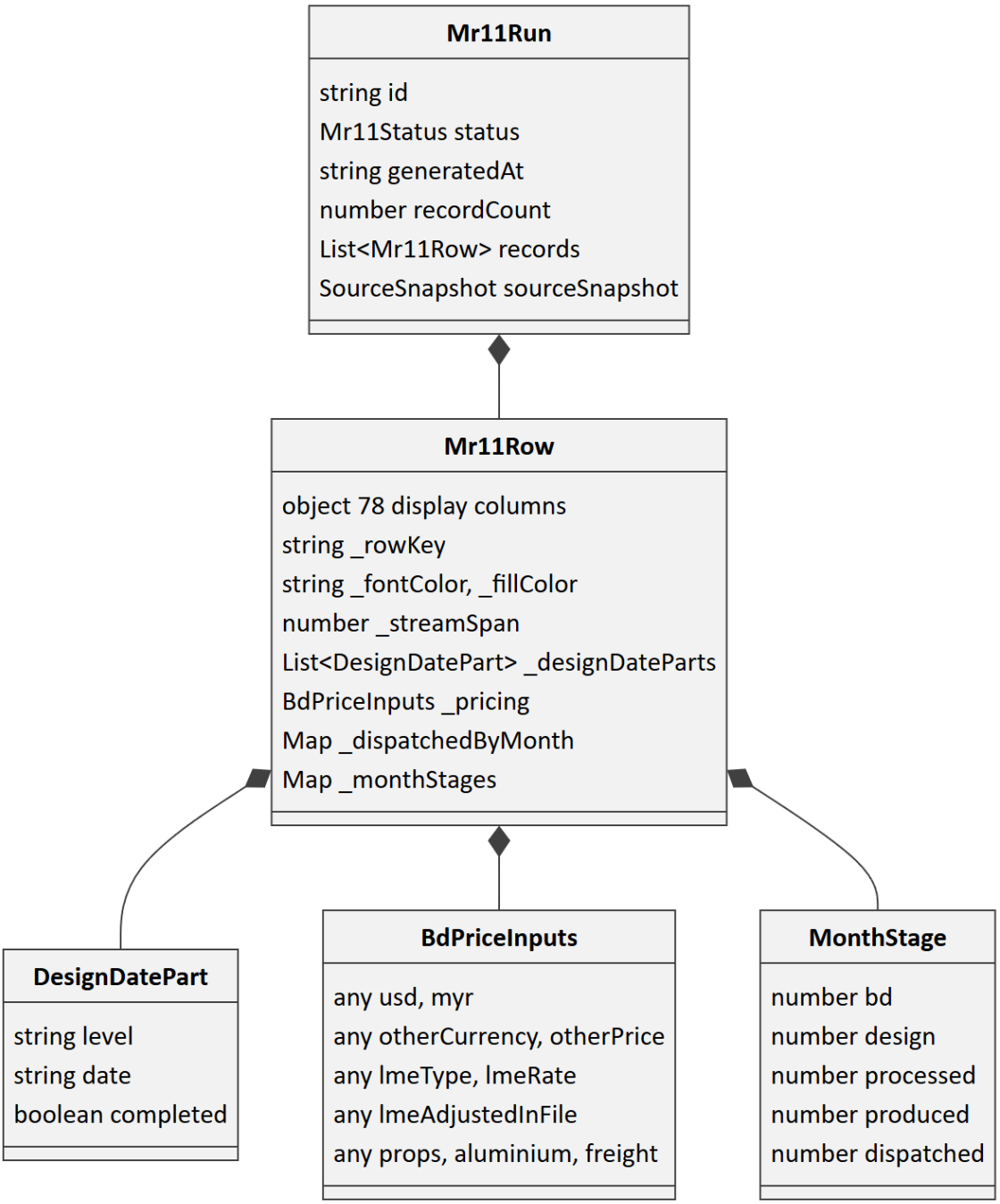


Figure 7: Mr11Run, its rows and each row's internal fields

6.1 An MR11 row

Each row is a JSON object holding the 78 columns shown on screen (listed with their types in Appendix A.5 of the System Architecture Document), plus some internal fields whose names start with an underscore.

Internal field	Type	What it means
_rowKey	Text	Identifies the row from one build to the next, using its short name, stream, font, fill and product type. It is used to delete and restore rows
_fontColor and _fillColor	Text	The BD row's colours, used on screen and in the export
_streamSpan	Number	On the first row of a block, how many rows share its Shell Plan and Design cells; 0 on the rows that follow
_designDateParts	A list of DesignDatePart	The completion dates for each level, as { level, date, completed }
_pricing	BdPriceInputs	BD's price inputs: usd, myr, otherCurrency, otherPrice, lmeType, lmeRate, lmeAdjustedInFile, props, aluminium and freight
_priceNote	Text	Explains a price that couldn't be converted, for example because the currency wasn't recognised
_dispatchedByMonth	A map from month (such as "2026-10") to m²	The m² dispatched in each month
_dispatchFile	LOCAL or OVERSEAS	Which Dispatch file the row uses
_monthStages	A map from month (such as "Oct-26") to MonthStage	The CEO pipeline for each month: bd, design, processed, produced and dispatched, all in m²
_designOnly	True or false	True for Re-Design Only rows, which are in blue font
_designDate, _dispatchedDate and _atdDate	Text holding a date	Single dates that the CEO Executive Hub uses to group figures by month

6.2 The source snapshot

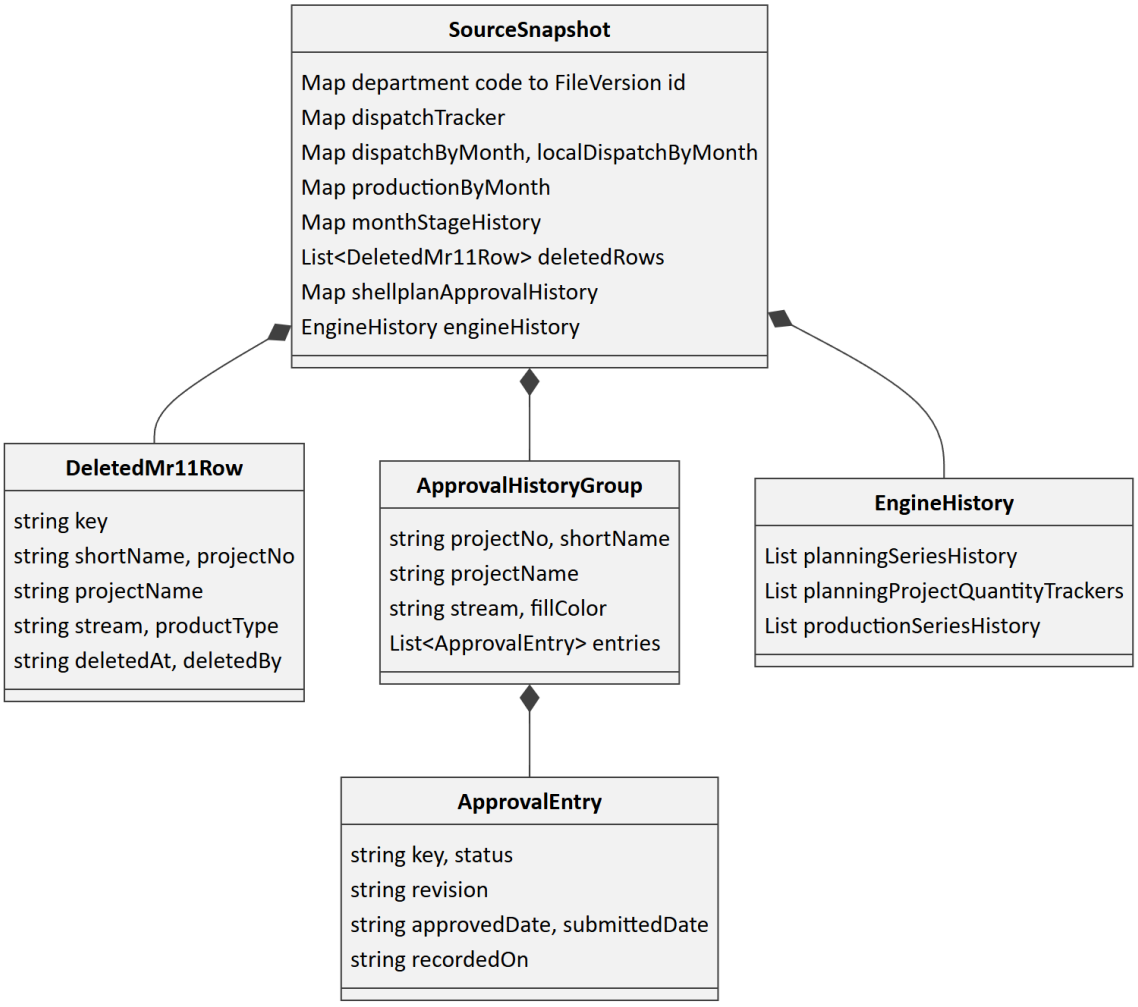


Figure 8: The source snapshot carried from one build to the next

Field	Type	What it means
BD, FINANCE, DESIGN and the other department codes	Text	The FileVersion id of the workbook each department contributed. If a newer upload exists, the run is out of date
dispatchTracker	A map from row key to { quantity, date }	The last Total Dispatch, and the date it changed
dispatchByMonth	A map from row key to a map from month to m²	The Overseas Total Dispatch at the last upload of each month
localDispatchByMonth	A map from row key to a map from month to m²	The Local m² dispatched in each month, kept after the monthly file resets
productionByMonth	A map from row key to a map from month to m²	The m² produced in each month, kept after the monthly file resets
monthStageHistory	A map from row key to a map from month to { design, processed }	The Design and Processed forecast for each month, as it last stood

Field	Type	What it means
deletedRows	A list of DeletedMr11Row	The rows an administrator removed, with their key, names, stream, product type, deletedAt and deletedBy
shellplanApprovalHistory	A map from stream key to ApprovalHistoryGroup	Every Approved and Reapprove seen so far, with its revision and dates
engineHistory	EngineHistory	The Planning series, the Planning quantity trackers and the Production series

7. API models

Every response uses the same envelope: `{ success, data }` when it works, and `{ success: false, error: { code, message } }` when it doesn't.

Model	Fields	Returned by
LoginResponse	token, and user { id, email, fullName, roles[], usingDefaultPassword }	POST /api/auth/login
CurrentUser	user { id, email, fullName, roles[] }	GET /api/auth/me
Department	id, code, name, description, activeVersionId, and activeVersion { id, originalFilename, status, uploadedAt, parsedWorkbook, uploadedBy }	GET /api/departments and /api/departments/:code
Mr11Response	run, fxRate, currencyRates[], lmePrice, visibleColumns, orderedHeaders, headerGroups, numberFormats, columnDepartments and departmentColors	GET /api/mr11
FxRate	rate, source, asOf and live	Part of Mr11Response
CurrencyRate	code, rate, inBd and source	Part of Mr11Response
LmePrice	cash, threeMonth, previousCash, asOf, source and live, in USD per tonne	Part of Mr11Response
HeaderGroup	label, and columns [{ key, label }]	Part of Mr11Response
PlanningSeries	projectNo, projectShortname, projectName, stream, fontColor, seriesNumber, totalProcessed, totalQuantity and closingDate	GET /api/mr11/planning-series
ProductionSeries	projectShortname, projectNo, projectName, stream, fontColor, seriesNumber and totalProduced	GET /api/mr11/production-series
ApprovalHistoryRow	projectNo, shortName, projectName, stream, fillColor, status, revision, approvedDate, submittedDate and recordedOn	GET /api/mr11/shellplan-history
DispatchMonthlyRow	shortName, projectNo, stream, productType, file, fontColor, fillColor, totalDispatch and months	GET /api/mr11/dispatch-monthly
DeletedRow	key, shortName, projectNo, projectName, stream, productType, deletedAt and deletedBy	GET /api/mr11/deleted-rows

Model	Fields	Returned by
AdminUser	id, email, fullName, createdAt and roles[]	GET /api/admin/users

8. Browser state

8.1 Who is signed in (the Zustand store in src/store/auth.store.ts)

Field	Type	What it means
token	Text, or nothing	The session token (a JWT)
user	An object, or nothing	{ id, email, fullName, roles[] }
usingDefaultPassword	True or false	True until the user changes the default password

8.2 What the browser remembers in localStorage

Key	Type	Set by	What it remembers
mfe_token	Text	The sign-in store	The session token, sent with every API call
mfe_user	JSON	The sign-in store	The signed-in user
mfe_default_password	"1"	The sign-in store	That the default password is still in use
sidebar.open	True or false	The sidebar	Whether the sidebar is expanded or collapsed
mr11.zoom	Number	The MR11 page	The table's zoom level, from 20% to 135%
mr11.sidePanelOpen	True or false	The MR11 page	Whether the Options panel is showing
mr11.hiddenColumnGroups	A list of text	The MR11 page	Which column groups are hidden
mr11.showRatesStrip	True or false	The MR11 page	Whether the market cards are showing
zoom.PLANNING, zoom.PRODUCTION and the other department codes	Number	The department pages	The sheet's zoom level on each department page
design.showApprovalHistory	True or false	The Shell Plan & Design page	Whether the approval history is showing
dispatch.part	LOCAL or OVERSEAS	The Dispatch page	Which Dispatch file is shown
dispatch.showMonthly	True or false	The Dispatch page	Whether the monthly dispatch table is showing

The MR11 filters and the search box are only kept while the page is open.

9. The in-memory store on the server

Each server instance keeps a working copy of the data in `server/db/prisma.ts`. It uses the same model names as the database, so the code reads as if it were using Prisma.

Collection	Key	Where its data comes from
users and userRoles	id	The Supabase User and UserRole tables (the seed accounts are used until these are loaded)
roles	id, such as role-admin	Built into the code
departments	id, such as dept-bd	Built into the code, with activeVersionId set from Supabase
fileVersions	id	The active versions in the Supabase FileVersion table
mr11Runs	A list	Builds made on this instance, or the latest one from Supabase
planningSeriesHistory	projectNo_stream_fontColor_seriesNumber	Mr11Run.sourceSnapshot.engineHistory
planningProjectQuantityTrackers	projectNo_stream_fontColor	Mr11Run.sourceSnapshot.engineHistory
productionSeriesHistory	projectShortname_stream_fontColor_seriesNumber	Mr11Run.sourceSnapshot.engineHistory
auditLogs and mr11Config	Not applicable	Memory only

10. Configuration models (server/config/mr11.config.ts)

Model	Fields	What it means
ColumnMapping	target, sourceDept, sourceColumn, type (string, number or date), exact and aliases	Which department column fills an MR11 column. "exact" means the whole header must match, and the aliases are older header names that are also accepted
ORDERED_HEADER_LIST	A list of text	The 78 MR11 columns, in the order they are shown
MR11_HEADER_GROUPS	A list of HeaderGroup	The columns that sit under a shared top header (Payment terms)
MR11_NUMBER_FORMATS	A map from column to number of decimal places	Fixed decimal places, such as 3 for LME Rate and 2 for payment amounts
MR11_USD_AMOUNT_COLUMNS	A list of text	The columns shown with a dollar sign
MR11_COLUMN_DEPARTMENTS	A map from column to department	Which department each column comes from, used for highlighting
MR11_DEPARTMENT_COLORS	A map from department to hex colour	Each department's highlight colour
MR11_SOURCE_KEY_MAP	A map from department to header names	The headers that identify the project in each department's file